

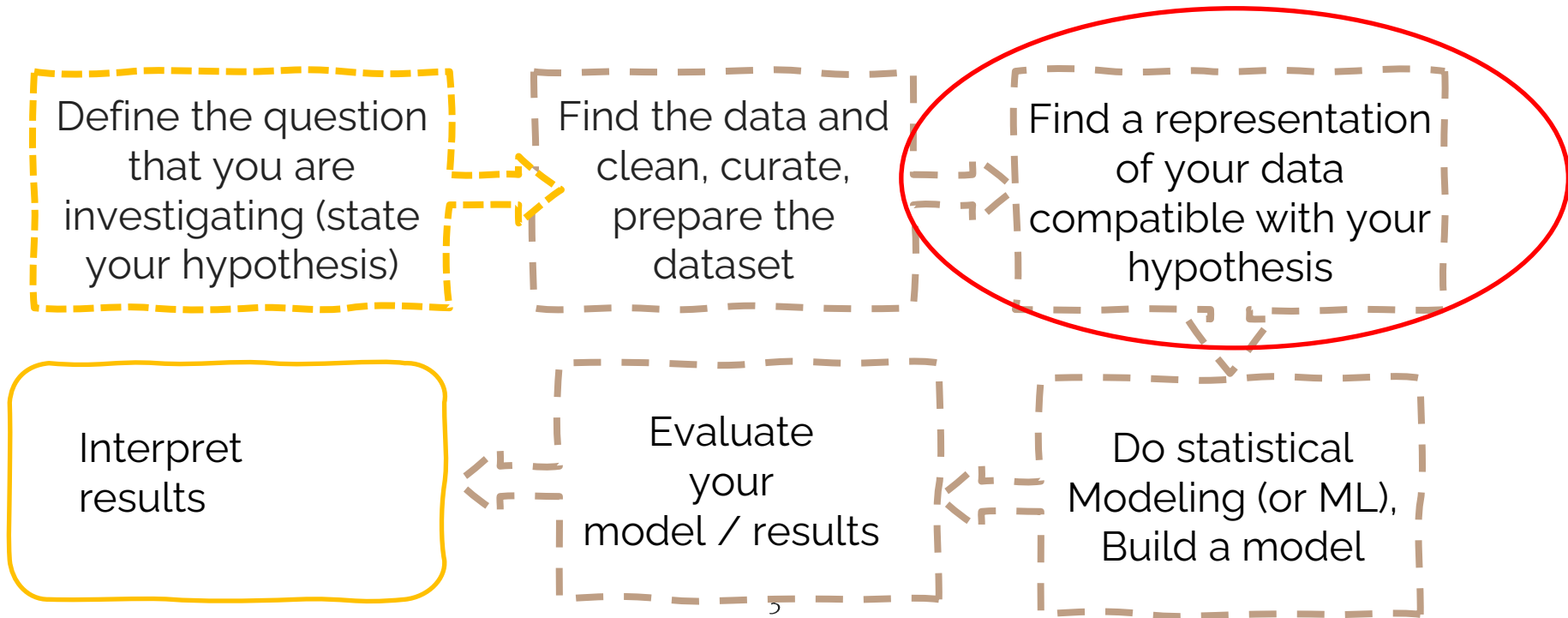
Introduction to programming. Music21

Anna Aljanaki

Kunstuniversität Graz

Computational musicology toolbox

Workflow



Software for computational analysis

- Music21 (Python-based)
 - musif
- MIRToolbox
- JSymbolic

JSymbolic

- Java software for feature extraction, part of jMIR toolbox
- Works with MIDI files
- Has both GUI and command line
- Can extract 246 different features
- Downsides
 - Not maintained since 2018
 - Is not meant for flexible feature addition

JSymbolic

- Pitch-related features (pitch histograms, number of distinct pitch classes)
- Melodic intervals (histograms, averages, medians)
- Chords
- Rhythm (note durations, inter-onset intervals, autocorrelation)
- Instrumentation
- Polyphonic structure
- Dynamics

JSymbolic

Symbolic 2.2
File Edit View Help

Information

SYMBOLIC FILES TO EXTRACT FEATURES FROM

File Name	File Path
F164_27_Tromboncino_Quanta_mai_cor.t.mid	C:\Users\Corv\Documents\Pu...
F164_28_Anon_Fraille_infelice_TransJS.mid	C:\Users\Corv\Documents\Pu...
F164_29_Anon_S_s_didi_Tril.mid	C:\Users\Corv\Documents\Pu...
F164_30_Anon_Qusi_foco_Tril.mid	C:\Users\Corv\Documents\Pu...
F164_31_Pesenti_So_ben_OMRCorriL.mid	C:\Users\Corv\Documents\Pu...
F164_32_Cara_Pesenti_Tromboncino_Quando_Is_OMRCorriL.mid	C:\Users\Corv\Documents\Pu...
F164_33_Pesenti_O_Dio_OMRCorriL.mid	C:\Users\Corv\Documents\Pu...
F164_34_Isaac_Hora_e_cord.mid	C:\Users\Corv\Documents\Pu...
F164_35_Compere_Che_fa_OMRCorriL.mid	C:\Users\Corv\Documents\Pu...
F164_36_Tromboncino_Che_debbio_OMRCorriL.mid	C:\Users\Corv\Documents\Pu...
F164_37_Obrecht_La_tortorella_OMRCorriL.mid	C:\Users\Corv\Documents\Pu...
F164_38_Josquin_Scarameilla_JRP_corL.mid	C:\Users\Corv\Documents\Pu...
F164_39_Fantini_Fantini_inesperata_OMRCorriL.mid	C:\Users\Corv\Documents\Pu...
F164_40_Anon_Jam_pis_OMRCorriL.mid	C:\Users\Corv\Documents\Pu...
F164_41_Anon_Donna Tu_corJ.mid	C:\Users\Corv\Documents\Pu...
F164_42_Pativano_Donne_vente_OMRCorriL.mid	C:\Users\Corv\Documents\Pu...
F164_43_Pativano_Un_cavaleri_OMRCorriL.mid	C:\Users\Corv\Documents\Pu...
F164_44_Festa_L_ultimo_di_OMRCorriL.mid	C:\Users\Corv\Documents\Pu...
F164_45_Anon_Vaghe_Is_OMRCorriL.mid	C:\Users\Corv\Documents\Pu...
F164_01_Pisano_Quanto_piu_OMRCorriL.mid	C:\Users\Corv\Documents\Pu...
F164_02_Pisano_Sl_e_debile_OMRCorriL.mid	C:\Users\Corv\Documents\Pu...
F164_03_Pisano_De_perche_OMRCorriL.mid	C:\Users\Corv\Documents\Pu...
F164_04_Pisano Son io i OMRCorriL_reversionJS.mid	C:\Users\Corv\Documents\Pu...

Add Files
Add Directory
Remove Files

Consistency Report
Contents Report
Play Sonification
Stop Sonification

FEATURES TO SAVE

Feature Name	Code	Values	MEI-Only
☒ Basic Pitch Histogram	P-1	128	No
☒ Pitch Class Histogram	P-2	12	No
☒ Folded Fifths Pitch Class Histogram	P-3	12	No
☒ Number of Pitches	P-4	1	No
☒ Number of Pitch Classes	P-5	1	No
☒ Number of Common Pitches	P-6	1	No
☒ Number of Common Pitch Classes	P-7	1	No
☒ Range	P-8	1	No
☒ Importance of Bass Register	P-9	1	No
☒ Importance of Middle Register	P-10	1	No
☒ Importance of High Register	P-11	1	No
☒ Dominant Spread	P-12	1	No
☒ Strong Tonal Centres	P-13	1	No
☒ Mean Pitch	P-14	1	No
☒ Mean Pitch Class	P-15	1	No
☒ Most Common Pitch	P-16	1	No
☒ Most Common Pitch Class	P-17	1	No
☒ Prevalence of Most Common Pitch	P-18	1	No
☒ Prevalence of Most Common Pitch Class	P-19	1	No
☒ Relative Prevalence of Top Pitches	P-20	1	No
☒ Relative Prevalence of Top Pitch Classes	P-21	1	No
☒ Interval Between Most Prevalent Pitches	P-22	1	No
☒ Interval Between Most Prevalent Pitch Classes	P-23	1	No
☒ Pitch Variability	P-24	1	No
☒ Pitch Class Variability	P-25	1	No

Select Default Features
Select All Features
Deselect All Features

PROCESSING INFORMATION

SUMMARY INFORMATION ON ALL IMPLEMENTED FEATURES:

- 245 unique features
- 1497 combined feature dimensions
- 228 unique one-dimensional features
- 18 unique multi-dimensional features
- 245 sequential features

Feature breakdown by type:

- 41 unique Overall Pitch Statistics features (190 total dimensions)
- 25 unique Melodic Intervals features (152 total dimensions)
- 35 unique Chords and Vertical Intervals features (183 total dimensions)
- 95 unique Rhythm features (449 total dimensions)

CONFIGURATION FILE AND WINDOWING SETTINGS

Load New Settings from a Config File
Save These Settings to a Config File

☒ Extract Features from Entire Files ☐ Extract Features from Windows

Window Duration (seconds):

Window Overlap Fraction (0.0 to 1.0):

FEATURE EXTRACTION AND SAVING SETTINGS

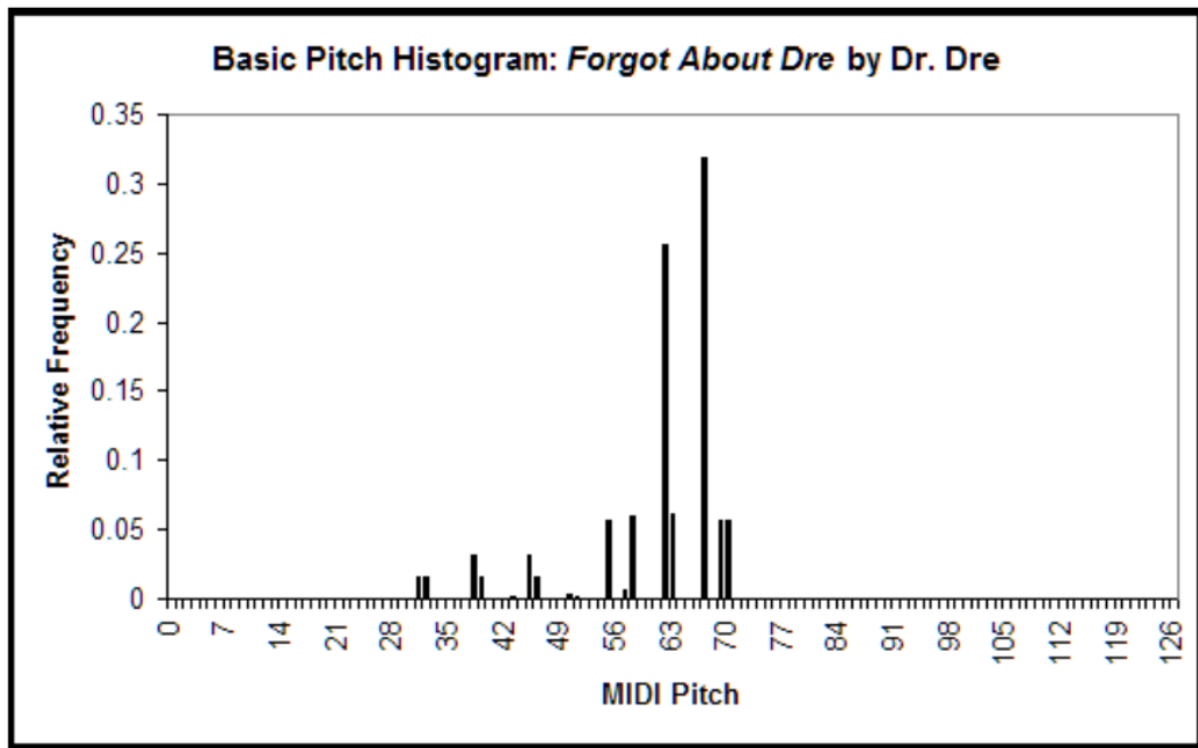
Set ACE XML Feature Values Save Path:

Set ACE XML Feature Definitions Save Path:

☒ Also Save Features in a Weka ARFF File ☒ Also Save Features in a CSV File

EXTRACT AND SAVE FEATURES

JSymbolic



MIDIToolbox

- A software for Matlab
- A counterpart of MIRToolbox
- Meter, melody, segmentation, similarity
- A bit more flexible than Jsymbolic because Matlab ecosystem allows it
- Downsides:
 - Matlab is proprietary
 - Latest version 2016

music21

- Python library for symbolic music analysis
- Corpora included with the library
- Works with MIDI and musicXML
- Most versatile, any possible features programmable
- Downside:
 - Does not provide pre-defined feature extraction pipelines
 - A lot of the code needs to be written ad hoc

Introduction to Python programming

For absolute beginners

What kind of snake is a python?

- Python is the most popular programming language worldwide (according to TIOBE index at least)
- It is a scripting language
 - Does not require compilation
 - Is interpreted
- Released in 1991

Why is it so popular?

- Simple syntax, similar to natural languages
- Versatile uses:
 - Software development
 - Scientific calculations
 - Machine learning
 - Web development
- Huge ecosystem of libraries

How can one run Python code?

- In a console with command prompts (.py files)
- Using IDE (PyCharm, VS Code)
- Using interactive environment (Jupyter) (.ipynb files)
 - Local installation (e.g. anaconda)
 - Colab notebooks

Opening a notebook

Google Colab

Let's you run code and store it on Google Drive

Requires Gmail-account

<http://colab.research.google.com/>

In case you do not have a Gmail account go to

mybinder.org